THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS UNIT TEST SET C

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Area of a rectangle of length 15 cm and breadth 9 cm is: 1

(a) 24 cm² **(b)** 150 cm² **(c)** 135 cm² **(d)** 48 cm²

2. The area of a figure is: 1

(a) the amount of surface enclosed by its boundary
(b) its longest side
(c) its angle
(d) the length of its boundary

3. $\frac{9}{11} - \frac{4}{11}$ equals: 1

(a) $\frac{4}{11}$ **(b)** $\frac{5}{0}$ **(c)** $\frac{13}{11}$ **(d)** $\frac{5}{11}$

4. Which is greater: $\frac{4}{7}$ or $\frac{5}{9}$? 1

(a) $\frac{4}{7}$ **(b)** cannot be compared
(c) both are equal **(d)** $\frac{5}{9}$

5. A compass keeps the pencil at a fixed distance, called the: 1

(a) chord **(b)** diameter **(c)** radius **(d)** arc

6. The longest chord of a circle is its: 1

(a) radius **(b)** tangent **(c)** arc **(d)** diameter

7. To construct a rectangle we need to know at least its: 1

(a) length and breadth **(b)** colour
(c) area only **(d)** weight

8. How many lines of symmetry does a regular hexagon have? 1

(a) 4 **(b)** 6 **(c)** 8 **(d)** 3

9. A parallelogram (not a rectangle) has: 1

(a) 2 lines **(b)** 4 lines
(c) no line of symmetry **(d)** 1 line

10. The image of a figure in a mirror is called its: 1

(a) reflection **(b)** rotation **(c)** area **(d)** translation

11. In the token model, a yellow token represents: 1

(a) −2 **(b)** −1 **(c)** +1 **(d)** 0

12. The absolute value (distance from zero) of −7 is: 1

(a) 14 **(b)** 7 **(c)** 0 **(d)** −7

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. Write the formula for the area of a triangle. 2

14. Define a proper fraction and an improper fraction with one example each. 2

15. State two properties of a rectangle. 2

16. Find the angle of rotational symmetry of a regular hexagon. 2

17. Find $(-8) + (-5)$. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. A right-angled triangle has sides 6, 8 and 10 cm. Find its perimeter and area. 3

19. Explain with a picture why $\frac{1}{2} = \frac{3}{6}$. 3

20. Describe how to draw the perpendicular bisector of a segment AB of length 7 cm. 3

21. State the rule for adding two integers with the same sign and with different signs. 3

22. Find the area of a rectangular floor 6 m by 4.5 m. Tiles of 50 cm × 50 cm (0.25 m²) cover it. How many tiles? 3

OR

Find the cost of fencing a rectangular field 30 m by 20 m at ₹15 per metre. 3

23. A rope 3 m long is cut into pieces of $\frac{3}{4}$ m. How many pieces? 3

OR

Show that $\frac{4}{6}$ and $\frac{10}{15}$ are equivalent. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) Compare $\frac{5}{6}$, $\frac{3}{4}$ and $\frac{7}{9}$. 5
(ii) Arrange in ascending order.

OR

(i) Define proper, improper and mixed fractions. 5
(ii) Write $\frac{29}{6}$ as a mixed fraction.
(iii) Write $4\frac{2}{7}$ as an improper fraction.

25. (i) Classify the letters A, B, E, H, N, S, Z, O by line symmetry. 5
(ii) Which have rotational symmetry?

OR

(i) Tabulate the lines and order of symmetry for rectangle, rhombus, parallelogram and kite. 5
(ii) Give reasons.

END OF QUESTION PAPER Total: 50 Marks

Mastermind